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PI Name:	Casner, Stephen Ph.D.		
Project Title:	Workload Tools and Guidelines		
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Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Space Human Factors Engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SHFH**:Space Human Factors & Habitability (archival in 2017)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
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Flight Program:			
Flight Assignment:	NOTE: Brian Gore became PI effective 10/01/2009, replacing Stephen Casner. See Gore for subsequent reporting (Ed., Oct 2011)		
Key Personnel Changes/Previous PI:	NOTE: Brian Gore became PI effective 10/01/2009, replacing Stephen Casner.		
COI Name (Institution):	Gore, Brian F. (NASA Ames Research Center)		
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Task Description:	This directed research project (DRP) will survey the available literature on: (1) workload measurement techniques; and (2) the effects of workload on operator performance. The first set of findings will be used to provide practitioners with a collection of simple-to-use workload measurement techniques along with characterizations of the kinds of tasks each technique has been shown reliably address. This will allow design practitioners to select and use the most appropriate techniques for the task(s) at hand. The second set of findings will provide practitioners with the guidance they need to design for appropriate kinds and amounts of workload across all tasks for which the operator is responsible. This guidance will help practitioners design systems and procedures that ensure appropriate levels of engagement across all tasks, and avoid designs and procedures that result in operator boredom, complacency, loss of awareness, undue levels of stress, or skill atrophy that can result from workload that distracts operators from the tasks they perform and monitor, workload levels that are too low, too high, or too consistent or predictable. Lastly, an empirical study will be conducted to evaluate the usefulness of the workload measurement and management guidance that we produce.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Although reliable techniques exist for measuring operator workload (Hill et al, 1992), surprisingly little attention has been directed toward the question of how workload affects performance in extended missions, particularly in extreme environments with expert operators such as those encountered in space operations. Classic human factors studies have demonstrated how prolonged periods of low workload can result in boredom and loss of awareness, while periods of high workload can lead to high error rates, narrowing of attention, and decreased awareness (Sheridan, 2002). The simplest interpretation of these results might lead one to conclude that workload is best maintained at an intermediate level, in which operators are neither pressed to their limits nor left to daydream. A number of researchers have proposed the idea of designing for an optimal level of workload in which the operator is always engaged but never overworked. Unfortunately, even this simple idea is wrought with complications. First, there is little reason to believe that any constant level of workload will avoid problems with boredom and decreased awareness. Second, studies such as Warm, Dember, and Hancock (1996) have challenged our simplistic notions of what tasks result in low and high workloads by demonstrating how seemingly low-workload vigilance tasks can in fact be “capacity-draining assignments” that can yield high workload measures. Third, other studies show that not all workload is created equally. A recent study by Casner (2009) showed that pilots’ navigational awareness varied with the amount of workload that was specifically devoted to the navigational task. Simply doing work does not seem to guarantee awareness, or freedom from boredom. A last challenge to the notion of an optimal level of workload is that there is simply no scientific support for the idea that working hard, then taking a break afterwards, is a bad thing. In fact, periods of high workload may allow operators to practice the multitasking skills needed in high-tempo situations such as emergencies. The takeaway message from the available research is that the relationship between workload and performance is a nuanced one. Simple approaches that attempt to maintain overall workload at an intermediate value are unlikely to achieve the desired results. The successful management or evaluation of workload must include a consideration of the individual tasks that operators must perform, the combinations of tasks that are performed during a work period, priorities among tasks, and individual differences among operators. The challenge to workload measurement and management is heightened when applied to extended missions, since this mission is outside of the micro-task environment for which many of the measurement methods and metrics were develop and have been applied. Two questions that motivate this DRP are (1) how do we apply the “micro”, individual level workload findings to the “macro” system level considerations, and (2) how is or should workload be managed across the integrated system? The first question deals with defining workload’s meaning in guiding successful system design. The second question raises the notion of coordinated/shared workload and whether there is a level of workload at which additional crewmembers are needed to insure successful performance. References: Hill, S. G., Iavecchia, H. P., Byers, J. C., Bittner, A. C., Zaklad, A. L., & Christ, R. E. (1992). Comparison of four subjective workload rating scales. Human Factors, 34(4), 429-439. Sheridan, Thomas B. (2002). Humans and Automation: System Design and Research Issues: John Wiley & Sons, Inc. Warm, J.S., Dember, W.N., & Hancock, P.A. (1998). Workload and vigilance. Paper presented at the Human Factors and Ergonomics Society 42nd Annual Meeting. Casner, S. M. (2009). Perceived vs. Measured Effects of Advanced Cockpit Systems on Pilot Workload and Error: Are Pilots' Beliefs Misaligned With Reality? Applied Ergonomics, 40(3), 448-456.
	An initial overview of measurement and workload evaluation techniques document was produced in FY09, termed the Workload Primer (WP). The WP focused on empirical data collection processes and culminated in the development of a “how-to” guide for some of the considerations in selecting a workload measurement approach for designs that are proposed to maximize human performance (e.g. display, interface, task, configuration, environment). The FY10-11 work will extend this to consider workload from a system perspective. As part of developing the WP, Casner & Gore (2008) developed a criterion to include articles into the collection of relevant workload articles that would be used. The criterion for inclusion was to limit the articles to only those that were peer reviewed, long standing and accepted in the field, applicable to a relevant range of conditions in a select domain of interest, with parallels being drawn in an attempt to identify analogous “extreme” environments to those in space. The goal was to start broad then refine. An initial review of over 225 articles was conducted. This list of articles was entered into EndNote. The EndNote reference list database includes titles, author list, abstracts, publication source, year, and includes hyperlinks to the .pdf article where available. The reference list included a range of conditions in the domain of interest (subj/obj measures), the seminal works in workload (e.g. Hart & Staveland, 1988 (TLX); Roscoe & Ellis, 1990 (Bedford); Cooper-Harper, 1969; Reid, 1985 (SWAT); etc) as well as summary works (e.g. Boff, Kaufman, Thomas, 1986). A down select from these 225+ articles was conducted to arrive at the final set of workload articles. All articles were reviewed and evaluated on uni-dimensional and multi-dimensional considerations. Casner & Gore also examined the notion of thresholds and the conditions that may benefit mostly from the various methodological approaches. Other

Task Progress:	considerations included whether the tools would be suitable for guiding a requirement-related and design-related question. The current workload toolset includes NASA TLX, Bedford, Cooper-Harper, Modified Cooper-Harper, Subjective Workload Assessment Technique (SWAT), Instantaneous Self Assessment of workload (ISA), and a host of physiological measures. The WP was used in one user test at NASA Ames Research Center, which led to modifications in the organization and the contents of the document. The results of the workload DRP studies completed in FY'09 attained 5 distinct aims: Aim A. A thorough understanding of workload measurement techniques and their applicability, ease of use, strengths and limitations for various human spaceflight tasks and activities was generated (Tasks 337-340). A review of techniques for measuring and evaluating designs with respect to Workload was also completed (Tasks 348-349). Aim B. The effects of workload on operator performance metrics (e.g. awareness, error, satisfaction) was highlighted and documented in the WP (Tasks 342, 343, 359). Aim C. Simple to use techniques that allowed for designs and operating procedures to be evaluated with respect to workload and operator performance was created and served to guide the development of the WP (Tasks 348-349). Aim D. The WP was developed to guide practitioners through the use of workload measurement techniques, task analysis techniques, and to explain the effects of varying degrees of workload on human operator performance (Tasks 358, 355, 354, 353). Aim E. The usability of the WP was tested by the Usability Team using a walkthrough methodology (Tasks 344, 360). References: Hart, S. G., & Staveland, L. (1988). Development of the NASA task load index (TLX): Results of empirical and theoretical research. In P. A. Hancock & N. Meshkati (Eds.), Human Mental Workload (pp. 239-250). Amsterdam: North Holland. Roscoe, A.H., & Ellis, G.A. (1990). A subjective rating scale for assessing pilot workload in flight: a decase of practical use (No. Technical Report TR 90019). Farnborough, UK: Royal Aerospace Establishment. Cooper, G.E., & Harper. R.P (1969). The use of pilot rating in the evaluation of aircraft handling qualities, NASA TN D-5153. April 1969. Reid, G.B., Potter, S.S., & Bressler, J.R (1989). Subjective workload assessment technique (SWAT): a user's guide (No. AAMRL-TR-89-023). WPAFB, Ohio: Armstrong Aerospace Medical Research Laboratory. Stein, E.S. (1985). Air traffic controller workload: An examination of workload probe. (Report No. DOT/FAA/CT-TN84/24). Atlantic City, NJ: Federal Aviation Administration Technical Center.
Bibliography Type:	Description: (Last Updated: 03/11/2010)
NASA Technical Documents	Casner SM, Gore BF. "Measuring and evaluating workload: A Primer." Final report NASA Ames Research Center to NASA Johnson Space Center. 2009. , Nov-2009